



Gram Complaint Management System

A scalable and modular **TypeScript + Express + MongoDB** backend for a complaint management system designed for **Villagers (Clients)** and **Gram Sevak (Admins)**.



Tech Stack

- **Node.js**
- **Express.js**
- **TypeScript**
- **MongoDB + Mongoose**
- **Zod (Validation)**



Project Structure

```
.
├── auth/                                # Node.js + Express Backend (Complaints)
│   └── src/
│       ├── config/                     # Database connection
│       ├── models/                     # Mongoose schemas
│       ├── schemas/                     # Zod validation schemas
│       ├── services/                   # Business logic
│       ├── controllers/                 # Request handlers
│       ├── routes/                     # API routes
│       ├── utils/                       # Helper functions
│       ├── app.ts                       # Express app setup
│       └── server.ts                   # Entry point
├── backend/                             # Python FastAPI (Microservice API Gateway)
│   └── app/
│       ├── api/                         # API Routing
│       ├── controllers/                 # Request handlers
│       ├── core/                       # Core configuration & startup
│       ├── middlewares/                 # Middleware functions
│       ├── model/                       # AI/NLP Models & logic
│       ├── services/                   # Business logic
│       ├── utils/                       # Helper functions
│       └── main.py                      # FastAPI entry point
└── frontend/                           # Next.js Web Application
    └── src/
        ├── app/                       # Next.js App Router (Pages & Layouts)
        ├── components/                 # Reusable UI components
        └── lib/                         # Utility functions & configs
```



Roles



Client (Villager)

- Register complaints (Text / Voice)
- Track complaint status using Complaint ID

Admin (Gram Sevak)

- View all complaints
 - Update complaint status
 - Manage resolution workflow
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Features

-  Complaint Registration
 -  Voice Complaint Support (Metadata Ready)
 -  NLP-based Categorization
 -  Complaint Status Tracking
 -  Status History Logging
 -  Modular Clean Architecture
-

Installation

```
git clone https://github.com/pd241008/Gram-Sevek.git
cd Gram-Sevek

npm install
```

Environment Variables

Create a `.env` file in root:

```
PORT=5000
MONGO_URI=your_mongodb_connection_string
```

Run the Server

Development

```
npm run dev
```

Production

```
npm run build
npm start
```

API Endpoints

🚀 Complaint Routes

+ Create Complaint

```
POST /api/complaints
```

Headers:

```
x-user-id: <USER_ID>
```

Body:

```
{
  "description": "Water supply issue in my area",
  "isVoice": false
}
```

📄 Get All Complaints

```
GET /api/complaints
```

🛠 Update Complaint Status (Admin)

```
PATCH /api/complaints/status
```

Headers:

```
x-user-id: <ADMIN_ID>
```

Body:

```
{
  "complaintId": "<DB_ID>",
  "status": "resolved"
}
```

🧠 NLP Categorization Logic

Complaints are automatically categorized based on keywords:

- "water" → Water Department
 - "road" → Road Department
 - "electricity" → Electricity Department
-

Database Design

Collections:

- **Users** (Client/Admin)
 - **Complaints**
 - **StatusHistory**
 - **NLP Logs (Optional)**
-

System Flow

Client → API → Validation → Service → Database → Response

Testing

Use tools like:

- Postman
 - Thunder Client
-

Future Enhancements

-  Authentication (JWT / OAuth)
 -  File Upload (Voice Storage)
 -  Notifications System
 -  Admin Analytics Dashboard
 -  Deployment (Docker + CI/CD)
-

Contribution

Contributions are welcome!

1. Fork the repository
 2. Create a new branch
 3. Commit your changes
 4. Submit a Pull Request
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Conclusion

This backend follows **clean architecture principles**, ensuring scalability, maintainability, and easy integration with any frontend (Next.js, React, Mobile Apps).

Contact

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